INTERNAL FEATURES OF HEART

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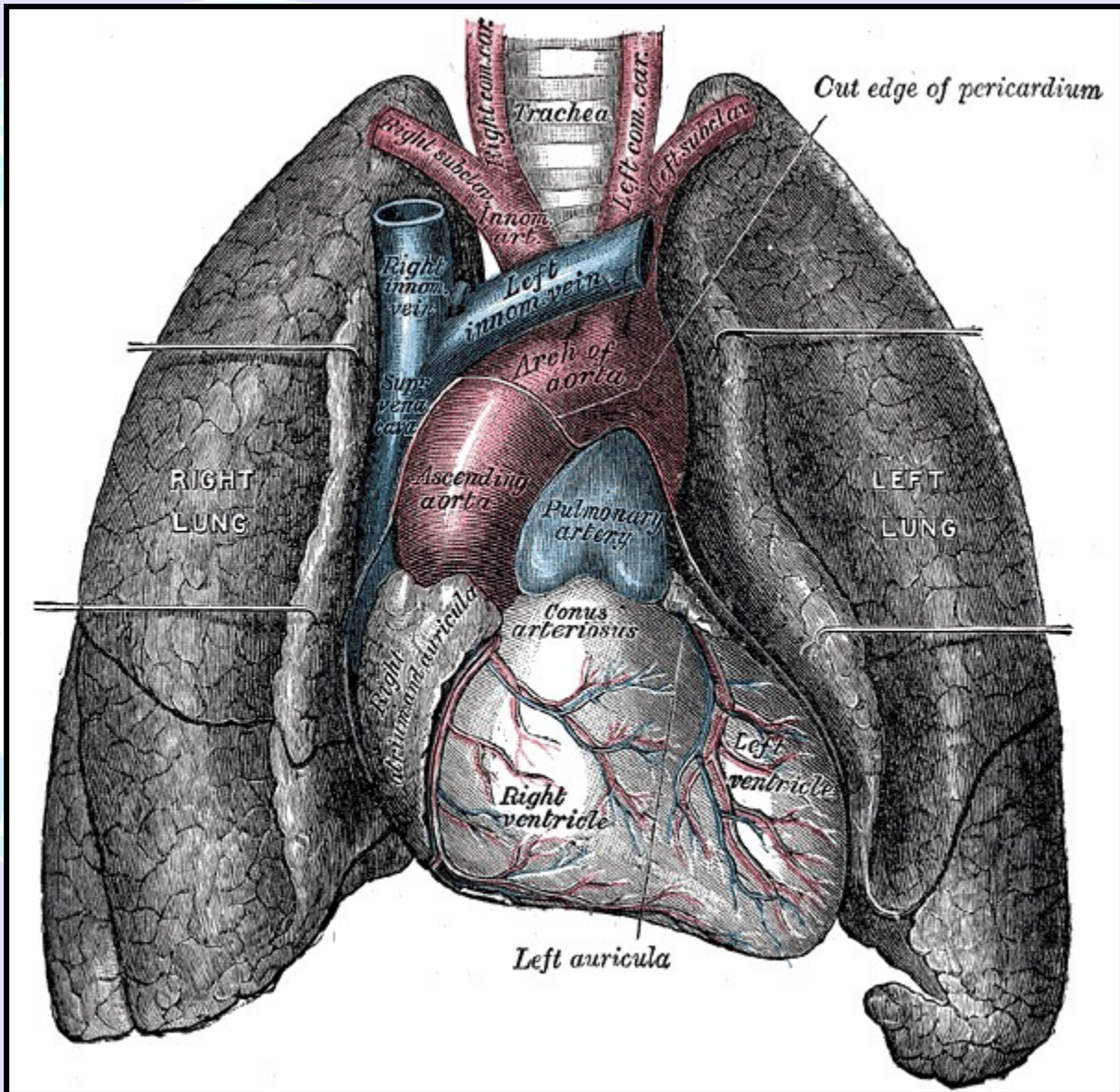
Learning objectives

At the end of the lecture the student should know the following

- Chambers of the heart
- Internal structures of the heart
- Valves of the heart

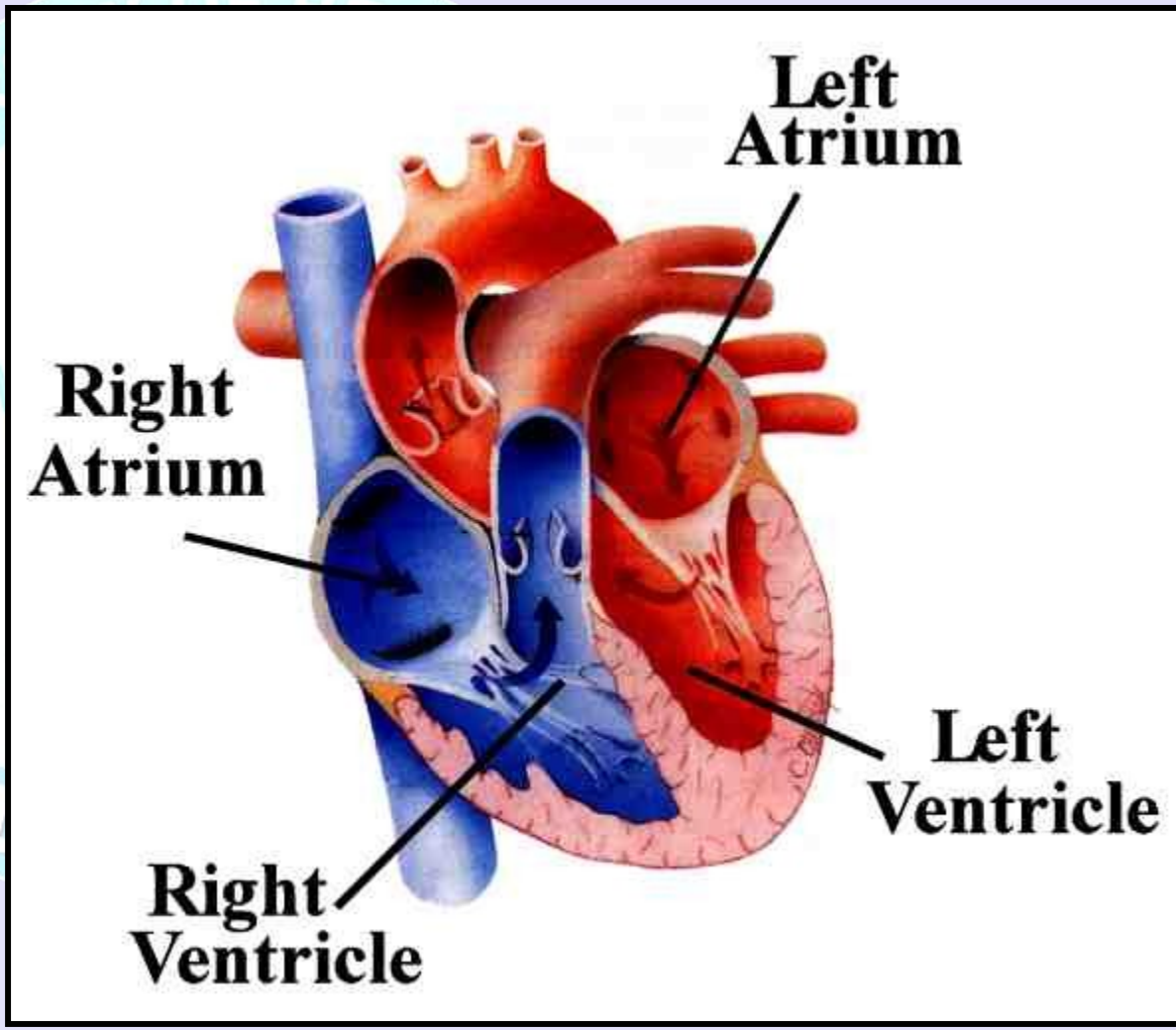
Heart

- Heart is a muscular organ.
- Lies between the lungs in the middle mediastinum
- One-third is on the right
- Two third in the left



CHAMBERS OF HEART

- Divided by septa into right and left halves
- Subdivided into atrium and ventricle
- So the heart has four chambers
- Right atrium
- Left atrium
- Right ventricle
- Left ventricle

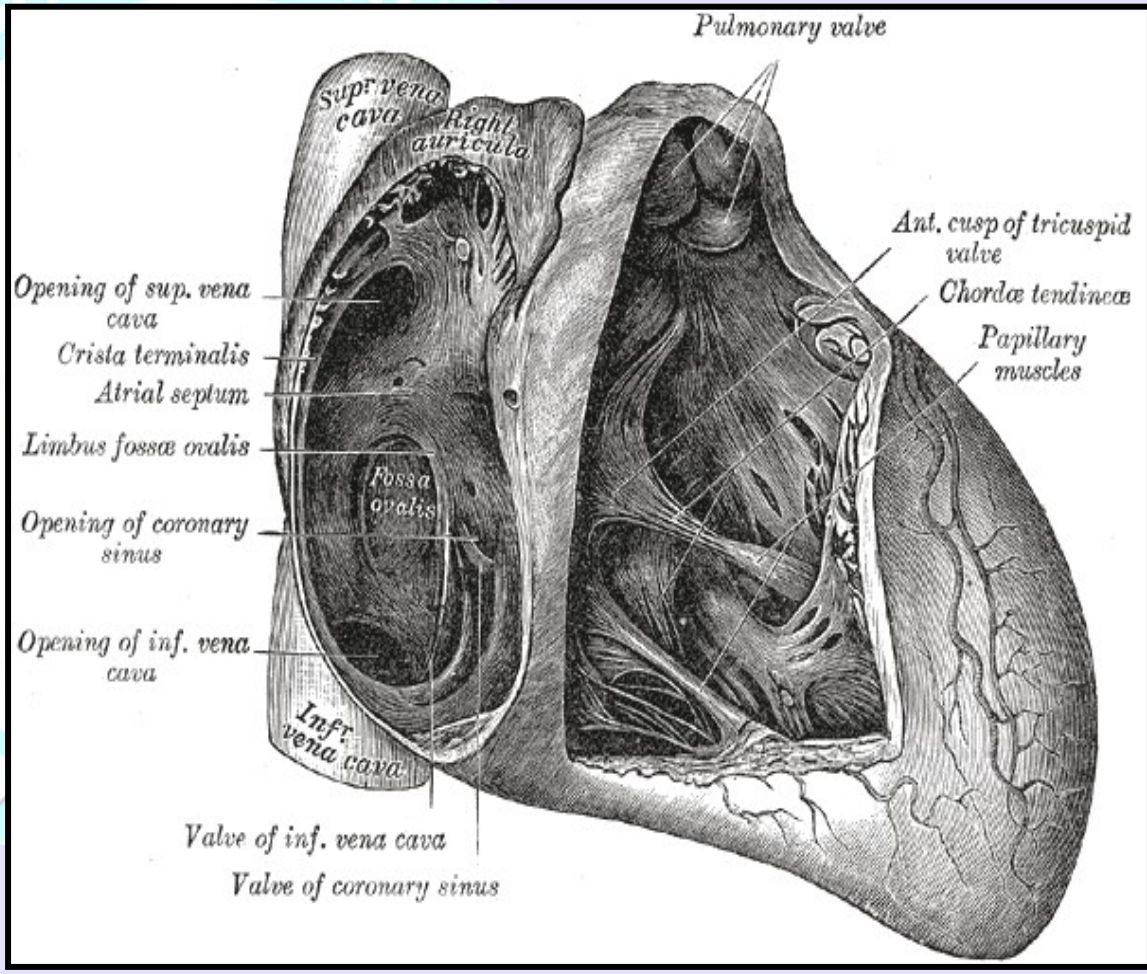


RIGHT ATRIUM

- Larger than left
- 57c.c capacity
- Consists of 2 parts

Main cavity called sinus venarum

Anterior small portion called auricula



Right atrium

- Rough ,muscular anterior wall composed of pectinate muscle.
- Smooth and rough part of atrial wall are separated by a shallow vertical groove the sulcus terminalis and internally by a vertical ridge– crista terminalis.

Openings of right atrium

- Superior vena cava

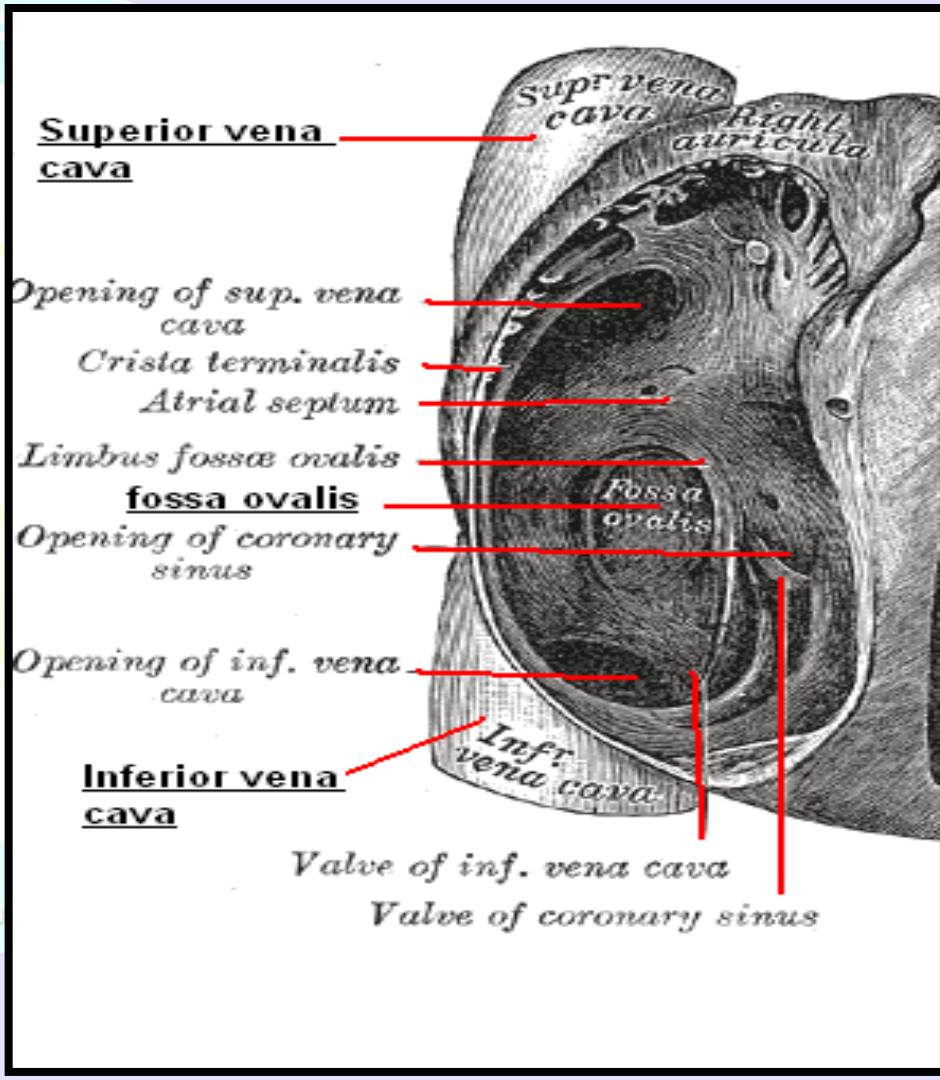
  Has no valve
- Inferior vena cava

  Eustachian valve
- Coronary sinus

  Btw the opening of inf. vena cava and AV opening

  Has a valve
- Foramina venarum

  Orifices of minute veins
- AtrioVentricular opening



Valves of right atrium

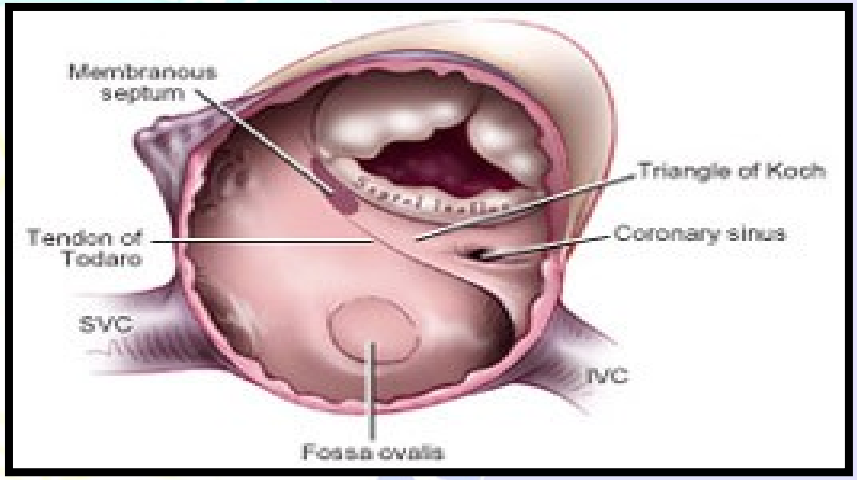
- **Valve of inferior vena cava**

  Semilunar valve
- **Valve of coronary sinus**

  Semicircular fold
- **Fossa ovalis**

  Oval depression on the septal wall of the atrium
- **Tricuspid valve**

  Between the right atrium and right ventricle



Right ventricle

- The right ventricle forms the largest part of anterior surface of heart, a small part of the diaphragmatic surface and almost entire inferior border of heart.

Right ventricle

- 85c.c capacity
- Extends from right atrium to near the apex of the heart
- Inferior border rests on diaphragm
- Anterosuperior surface is post sternal
- Interventricular septum makes the Posterior surface.

Right ventricle

- The interior of right ventricle has irregular muscular elevation --- **Trabeculae carneae.**
- A thick muscular ridge –supraventricular crest separates muscular wall of inflow part of chamber from the smooth part of outflow part.

Openings of right ventricle

- **Right atrioventricular orifice**

  Communication between right atrium and ventricle

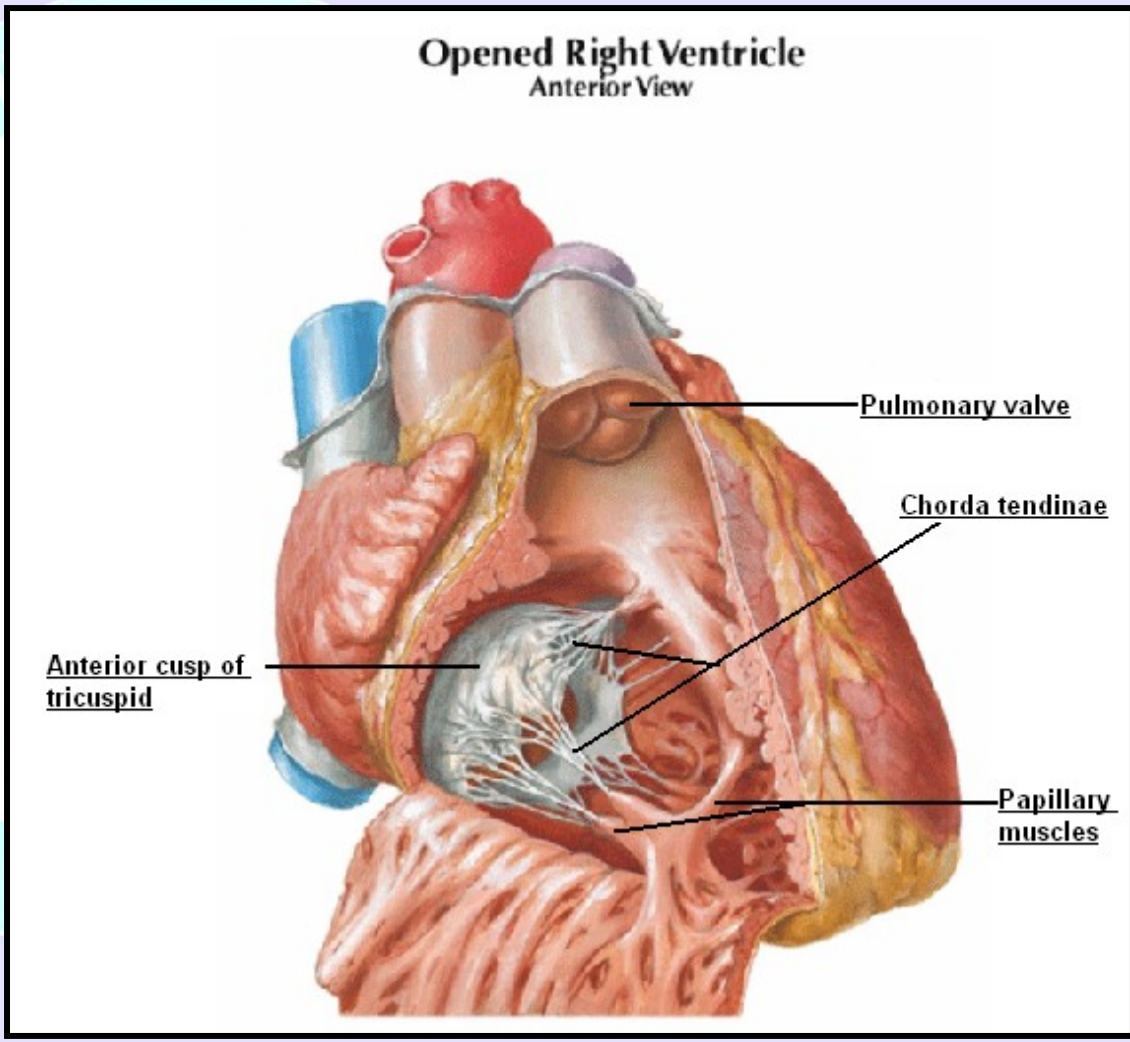
  4cm diameter

  Has a tricuspid valve

- **Pulmonary orifice**

  Circular in form

  Guarded by the pulmonary semilunar valves



Valves of right ventricle

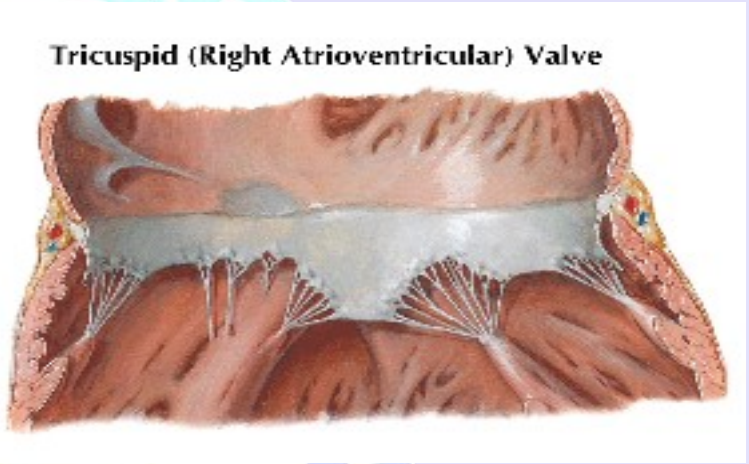
- **Tricuspid valve**

  Has 3 triangular cusps:

  Anterior or Infundibular cusp

  Posterior or Marginal cusp

  Medial or Septal cusp



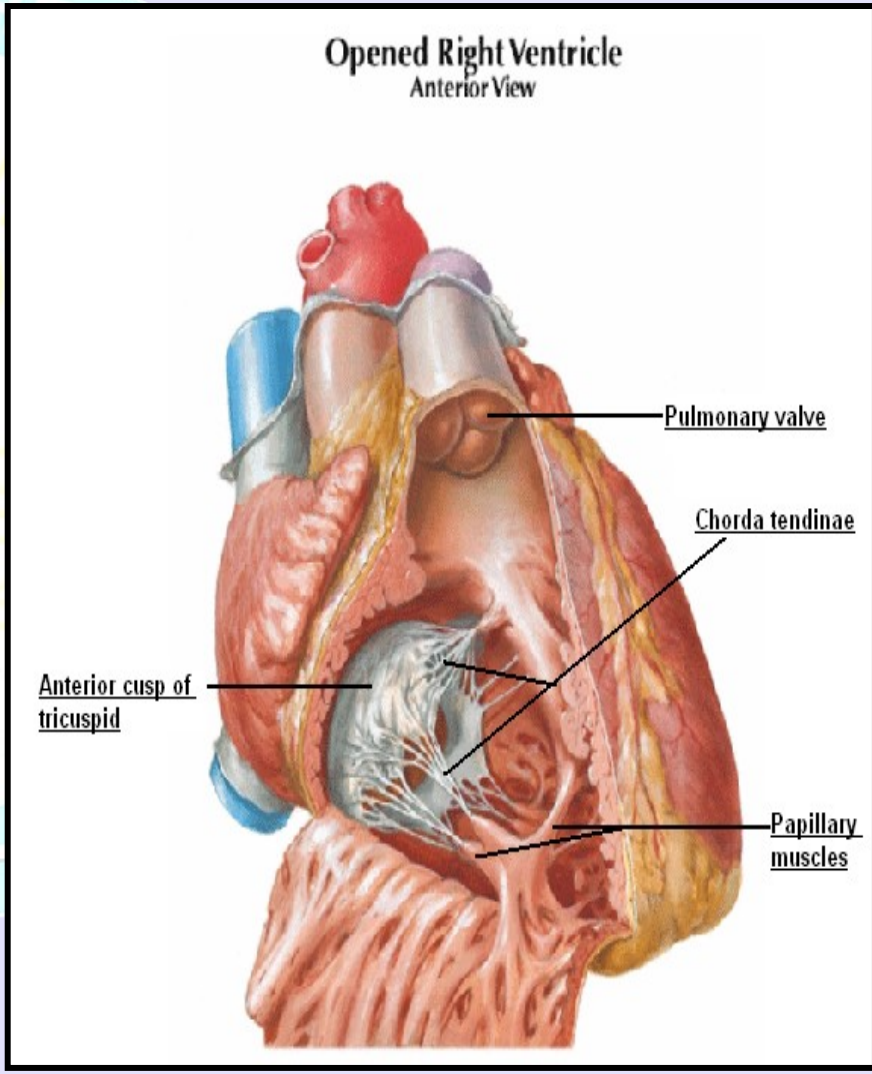
- **Pulmonary valve**

  3 in number

  Attached to the wall of artery at the junction of artery and ventricle

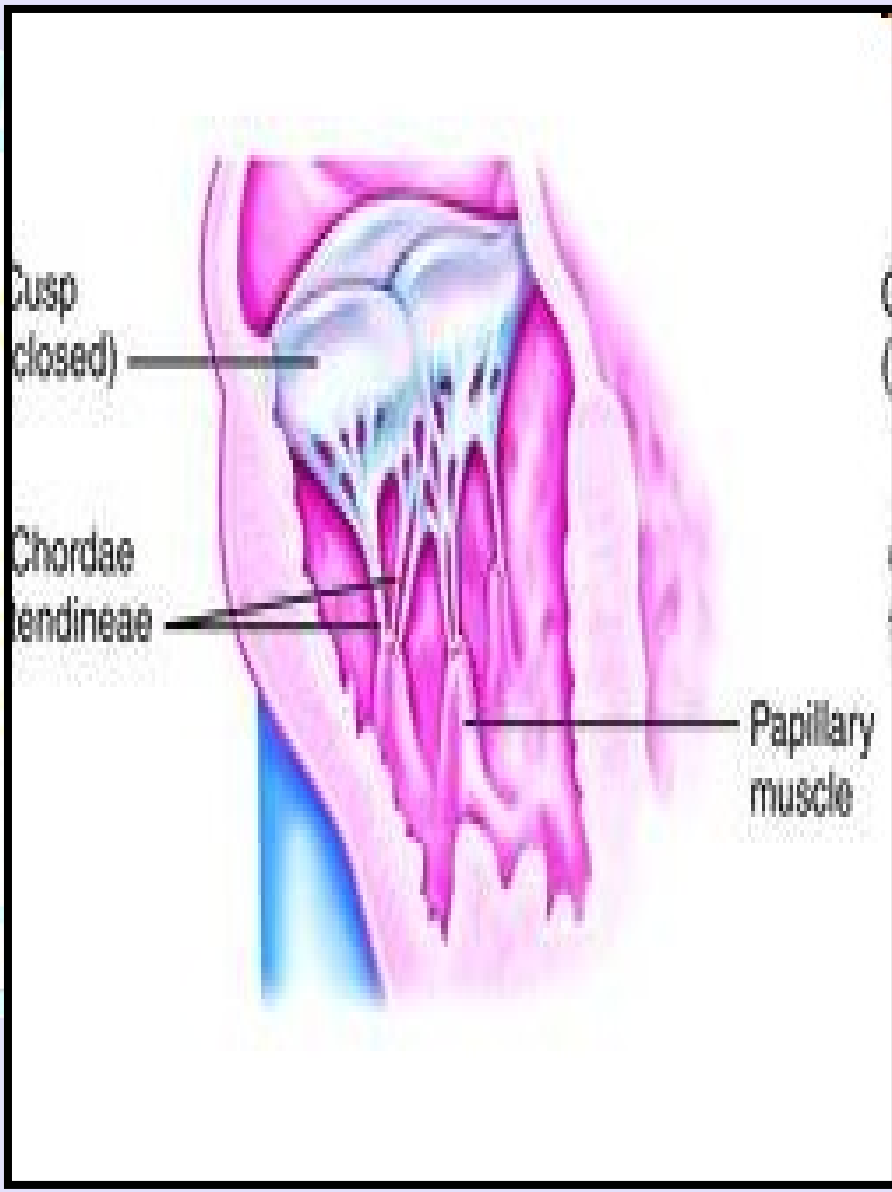
Tricuspid valve

- The bases of valve cups are attached to the fibrous ring.
- **Tendinous cords**: free edge and ventricular surfaces of anterior posterior and septal cups.
- These cords arises from the **papillary muscles** : are conical muscular projection – bases attached to the ventricular wall.



Tricuspid valve

- The three papillary muscles—correspond to the cups of tricuspid valve:
- **Anterior papillary muscles**: largest and most prominent, arises from the anterior wall of the vent. Its cords attach to the ant. And post.cups of valve.
- **Posterior papillary muscles**: smaller and may consist of several parts, arises from theinferior wall of right ventricle. Its cords attach to the posterior and septal cups.
- **Septal papillary muscles**: arises from the interventricular septum,its cords attach to the anterior and septal cups.

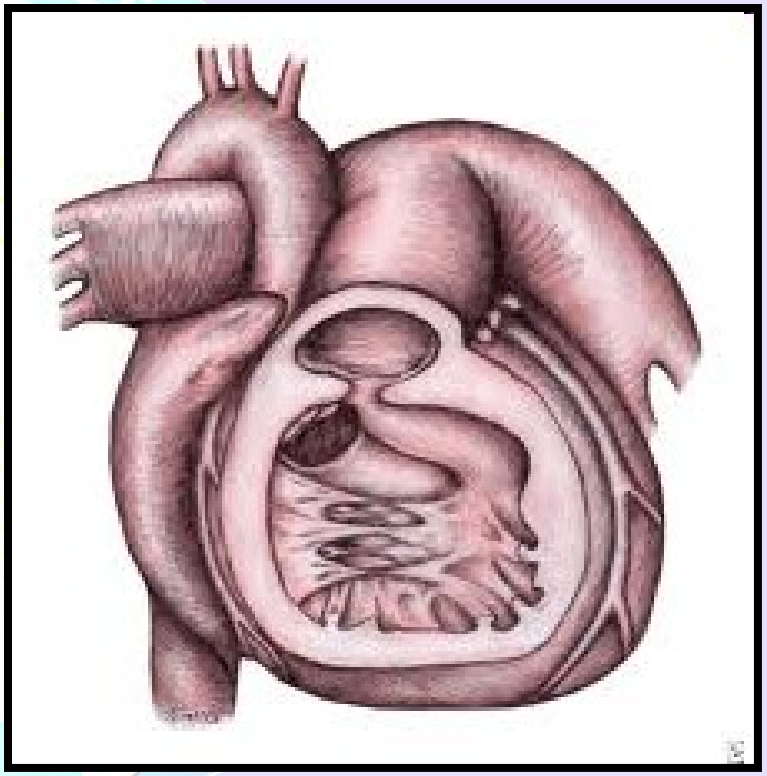


Interventricular septum

- Composed of muscular and membranous part b/w right and left ventricles..
- **Muscular part of the IVS** which form the majority of septum.
- The thin membrane a part of fibrous skeleton forms the much smaller **membranous part of IVS.**
- **Septomarginal Trabecula:**muscular bundle that traverses the right ventricular chambr from the inferior part of IVS to the base of ant.papillary muscle—right branch of AV bundle.

Pulmonary valve

- Pulmonary valve at the apex of conus arteriosus at the level of 3rd costal cartilage.



BLOOD FLOW IN RIGHT VENTRICLE

- When atrium contracts---Rt. Ventricle is empty and relaxed: blood –forced thru AV orifice—inflow of blood in rt. Ventricle posteriorly: when ventricle contract---blood flows into the pulmonary trunk.
- Blood flow is U shaped in direction: is accommodated by **supraventricular ridge**



THE END !!!!!